

Executive MBA

Financial Accounting

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Lecture 3: Operating Assets and Liabilities



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Topics

- *Current* operating assets and operating liabilities:
 - Financial reporting for inventory, the valuation of inventory, and the accounting for inventory in merchandising & manufacturing firms.
 - Financial reporting for contingent liabilities such as warranties and restructuring liabilities.
 - Provisions and earnings quality.
- *Non-current or long-lived* operating assets.

Inventory Accounting

Three key issues for inventory accounting:

1. What types of costs are capitalized in the acquisition of inventory?
2. What assumptions are used to trace the movement of costs into and out of inventory, *including* the effects of changes in per-unit inventory costs over time?
3. Should Inventory be marked to account? How to account for the impact of changes in the market value of inventories after acquisition?

Yamaha Corporation: Fiscal 2024 Inventories

Yamaha Corporation and its consolidated subsidiaries As of March 31, 2024 and 2023	Millions of yen		Thousands of U.S. dollars (Note 2)
	2024	2023	2024
Assets			
Current assets			
Cash and cash equivalents (Notes 7 and 34)	¥101,587	¥103,886	\$670,940
Trade and other receivables (Notes 8, 24 and 34)	88,015	75,392	581,302
Other financial assets (Notes 14 and 34)	4,861	1,089	32,105
Inventories (Note 9)	164,149	153,671	1,084,136
Other current assets (Note 10)	10,733	12,645	70,887
Total current assets	369,346	346,685	2,439,377

(5) Inventories

Inventories are measured at the lower of cost and net realizable value.

The cost of inventories is determined principally based on the weighted-average method and includes all costs of purchase, costs of conversion and other costs incurred in bringing the inventories to their present location and condition. Net realizable value is the estimated selling price in the ordinary course of business less the estimated costs of completion and the estimated costs necessary to make the sale, taking into account the salability.

9 Inventories

The breakdown of inventories is as follows:

	Millions of yen		Thousands of U.S. dollars (Note 2)
	2024	2023	2024
Finished goods and merchandise	¥109,670	¥96,890	\$724,325
Work in process	15,055	16,572	99,432
Raw materials and supplies	39,422	40,208	260,366
Total	¥164,149	¥153,671	\$1,084,136

Note: The amounts of write-downs (reversal of write-downs) of inventories recognized in "Cost of sales" for the fiscal years ended March 31, 2024 and 2023 were ¥4,097 million (\$27,059 thousand) and ¥1,300 million, respectively.

Harley Davidson: Fiscal 2023 Inventories

(in thousands)

	2023	2022
<u>ASSETS</u>		
Cash and cash equivalents	\$ 1,533,806	\$ 1,433,175
Accounts receivable, net	267,200	252,225
Finance receivables, net of allowance of \$67,035 and \$62,488	2,113,729	1,782,631
Inventories, net	929,951	950,960
Restricted cash	104,642	135,424
Other current assets	214,401	196,238
Current assets	5,163,729	4,750,653

Inventories, net – Substantially all inventories located in the U.S. are valued using the last-in, first-out (LIFO) method. Other inventories totaling \$447.5 million and \$425.0 million at December 31, 2023 and 2022, respectively, are valued at the lower of cost or net realizable value using the first-in, first-out (FIFO) method.

Inventories, net consisted of the following as of December 31 (in thousands):

	2023	2022
Raw materials and work in process	\$ 389,221	\$ 331,380
Motorcycle finished goods	514,964	549,041
Parts and accessories and apparel	150,844	187,039
Inventory at lower of FIFO cost or net realizable value	1,055,029	1,067,460
Excess of FIFO over LIFO cost	(125,078)	(116,500)
	\$ 929,951	\$ 950,960

Inventory obsolescence reserves deducted from FIFO cost were \$110.2 million and \$84.6 million as of December 31, 2023 and 2022, respectively.

Inventory Terminology

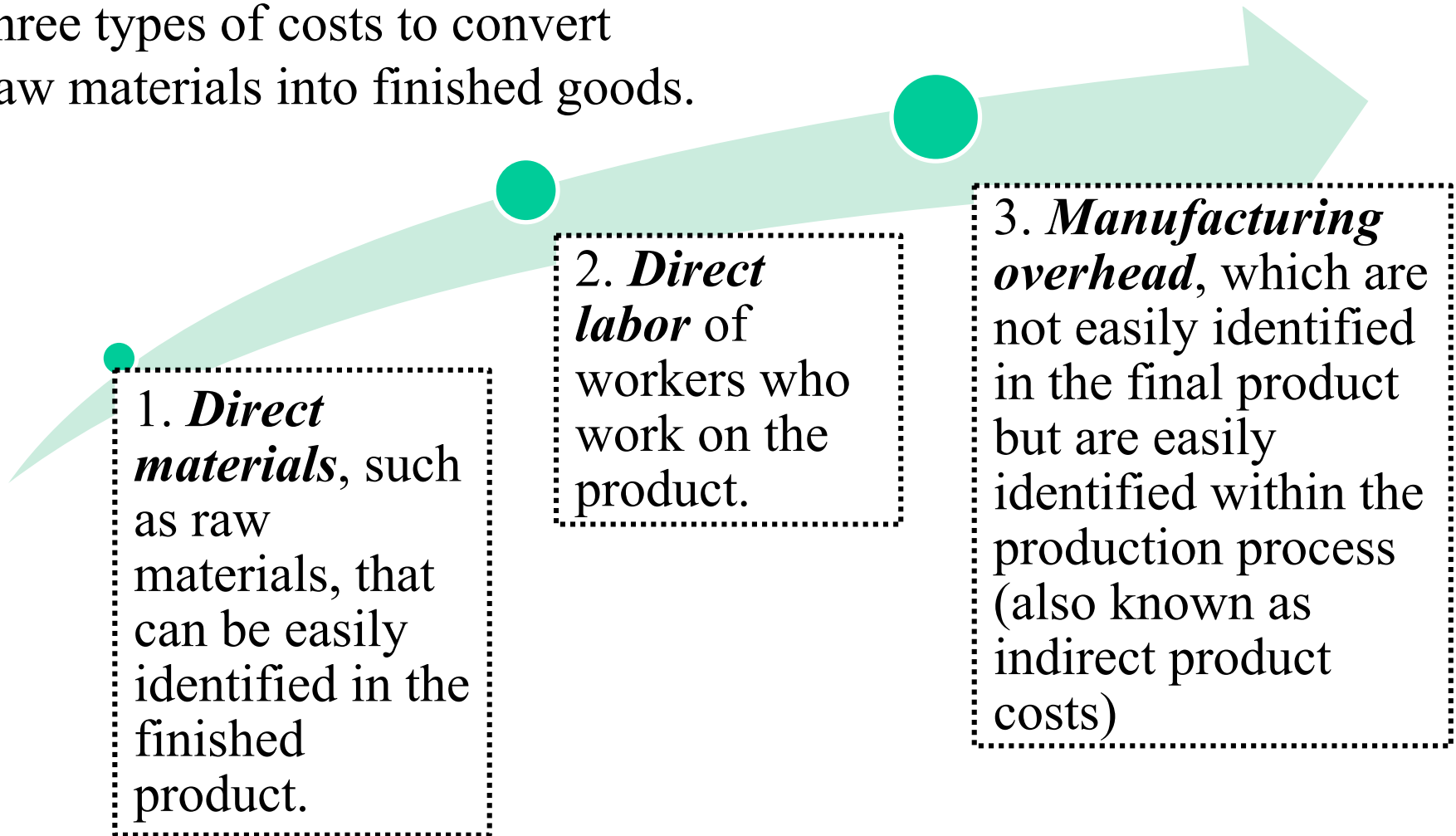
- ***Inventory***: stock of goods held by a firm either for sale or for further processing.
- ***Merchandise inventory***: goods held for sale by a retail or wholesale business.
- ***Finished goods inventory***: goods held for sale by a manufacturing firm.
- ***Raw materials inventory***: materials stored that are used in production.
- ***Work-in-process/progress***: inventory of *partially* completed goods.
- ***Cost of goods sold (COGS)***: denotes the transfer of costs from the Finished Goods Inventory account to an ***expense account***, reducing net income and ultimately retained earnings.

Which purchase costs are included merchandise inventory?

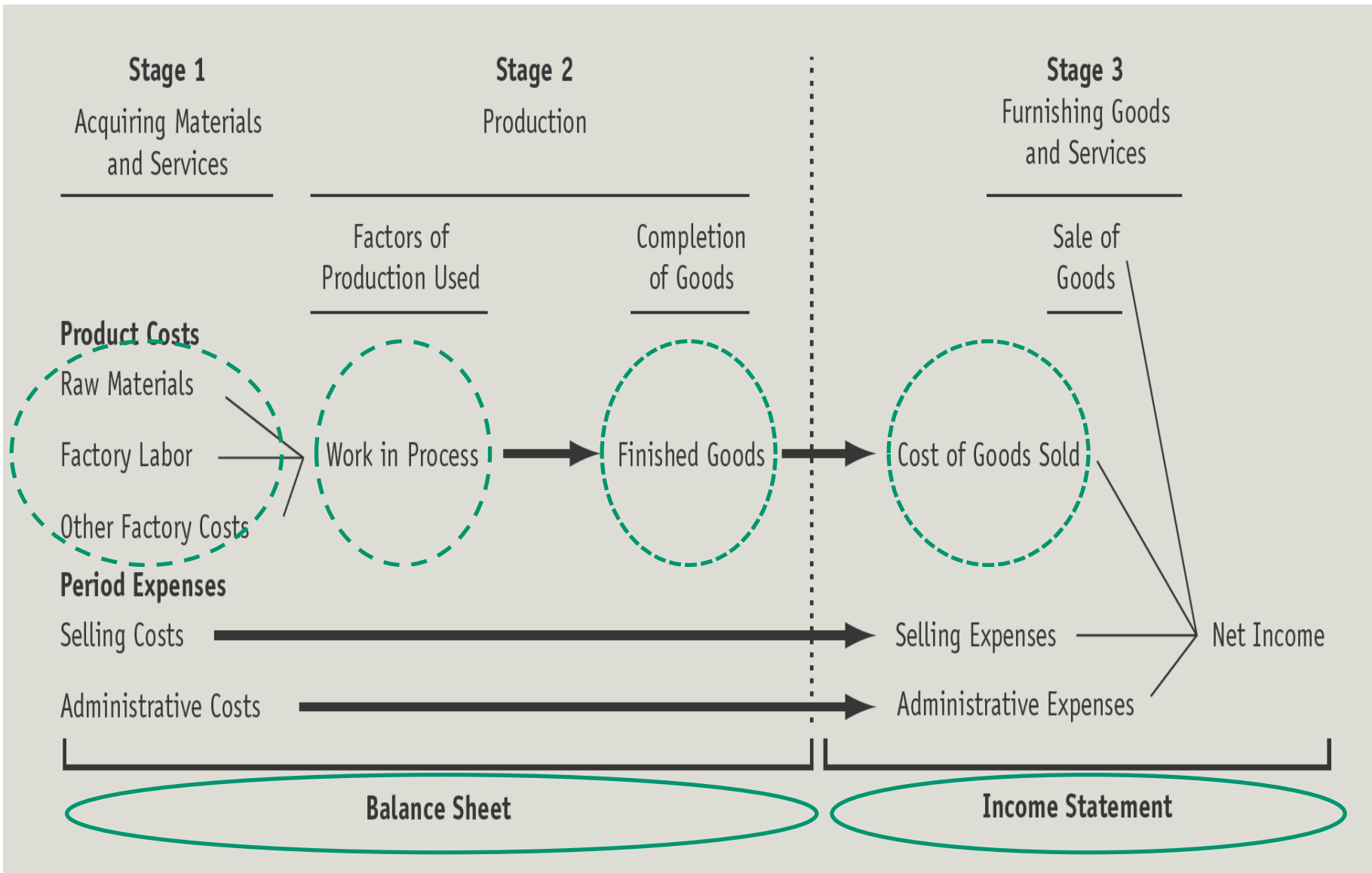
- Inventory should include all costs incurred to acquire goods and prepare them for sale:
 - *The purchase price* includes the cost of ordering, receiving, and inspecting goods and recording the purchase. The transaction is recorded when title passes to the firm.
 - *Merchandise purchase adjustments*, such as transportation and special handling costs are *capitalized* as part of costs included in inventory.

Which costs should be included in manufacturing inventory?

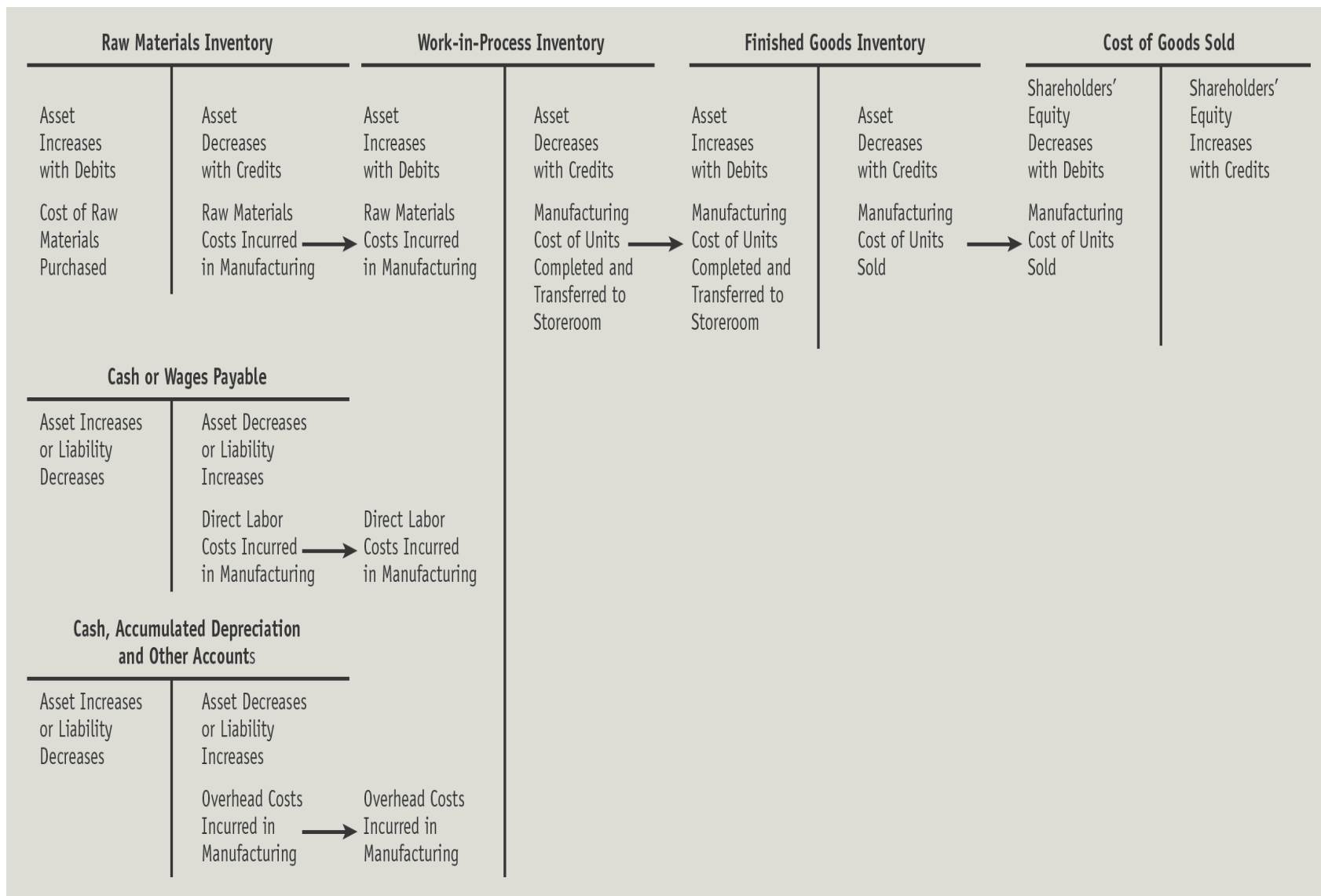
A manufacturing firm incurs three types of costs to convert raw materials into finished goods.



Cost Flows



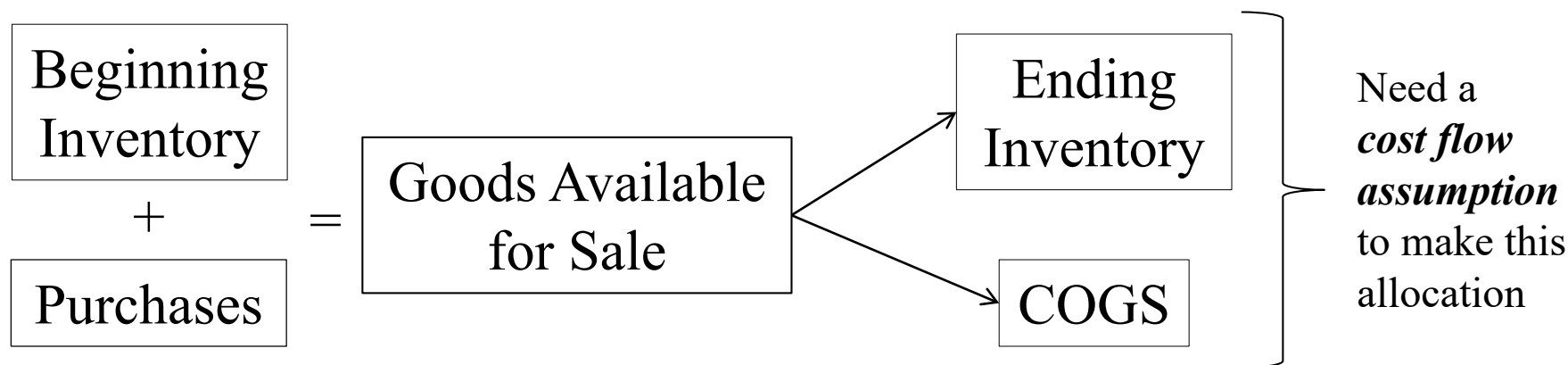
Flow of Manufacturing Costs through Accounts



Summary: What Costs Should Be Included in Inventory?

- All costs incurred to acquire and prepare inventory for sale:
 - Merchandising firms (e.g., Walmart): include unit price, freight charges, unpacking and inspections costs, less any discounts or returns.
 - Manufacturing firms (e.g., BMW): include raw materials, direct labor, and manufacturing overhead.
 - Raw materials, direct labor, and manufacturing overhead are called **product costs** as they are *capitalized* to the cost of inventory and are not expensed.
 - If overhead activities cannot be tied to production, they are assigned to SG&A or another income statement expense line and are thus **period costs**; e.g., advertising expenditures, office employees, etc.

Cost Flow Assumptions



- Neither U.S. GAAP nor IFRS requires firms to use the *specific identification method*.
 - Both set of standards allow firms to select a **cost-flow assumption**. More importantly, the cost-flow assumption selected *need not* match the actual physical flow of units within the firm.

Four Types of Cost Flow Assumptions

1. Specific Identification (SI)
2. Weighted Average (WA)
3. First-In, First-Out (FIFO)
4. Last-In, First-Out (LIFO): allowed under US GAAP only, not IFRS
 - Many US multinational companies use LIFO for their domestic subsidiaries only but FIFO and average cost for foreign subsidiaries.

Cost Flow Assumptions: A Simple Example

Assumed Data

Beginning Inventory:	TV Set 1 Cost		\$200
Purchases:	TV Set 2 Cost		400
	TV Set 3 Cost		450
Cost of Goods Available for Sale			<u>\$1,050</u>
Sales:	TV Set 2 Price		<u>800</u>

	Cost Flow Assumptions			
	SI	WA	FIFO	LIFO
Financial Statement	(1)	(2)	(3)	(4)
Ending Inventory	650	700	850	600
Sales	800	800	800	800
COGS	400	350	200	450
Gross Profit	400	450	600	350

Tradeoff between FIFO vs. LIFO

Cost Flow Assumption	Income Statement	Balance Sheet	Inventory-on-Hand Assumption
FIFO	Old Purchase Prices	Current Purchase Prices	Last-in-still-here
LIFO	Current Purchase Prices	<u>Very</u> Old Purchase Prices	First-in-still-here

- Note that, when input costs are *rising*, LIFO income is *lower* than FIFO income as long as inventories remain constant or are increasing.
- However, if inventories decrease, then LIFO income may be much higher than FIFO income. This phenomenon is known as *LIFO liquidation*.

LIFO Reserves

- LIFO usually leads to a balance sheet inventory value that is typically far below the current cost of that inventory.
- To allow meaningful comparison between LIFO and FIFO firms, **SEC registrants** using LIFO are required to disclose an amount called **LIFO Reserve**, which is the excess of current cost or FIFO cost over LIFO cost of inventories at each balance sheet date.
 - $\text{Inventory}_{\text{FIFO}} = \text{Inventory}_{\text{LIFO}} + \text{LIFO Reserve}$
 - The value of the LIFO reserve can be viewed as a *cumulative unrealized holding gain* due to rising inventory costs.
 - Such cumulative unrealized gains would be recognized as earnings via a lower COGS amount when the inventory is eventually sold.

Relationship between LIFO Reserve and COGS

$$Inventory_{FIFO} = Inventory_{LIFO} + LIFO \text{ Reserve}$$

$$\Delta Inventory_{FIFO} = \Delta Inventory_{LIFO} + \Delta LIFO \text{ Reserve}$$

$$Purchases - COGS_{FIFO} = Purchases - COGS_{LIFO} + \Delta LIFO \text{ Reserve}$$

$$COGS_{LIFO} = COGS_{FIFO} + \Delta LIFO \text{ Reserve}$$

- If a company *increases* its LIFO reserve during a year, so that $\Delta LIFO \text{ Reserve} > 0$, then $COGS_{LIFO} > COGS_{FIFO}$, resulting in *lower earnings* under LIFO relative to FIFO.
- If the company *dips* into its LIFO reserves, i.e., $\Delta LIFO \text{ Reserve} < 0$ (e.g., by slowing down its purchases relative to its sales) then $COGS_{LIFO} < COGS_{FIFO}$, resulting in a boost to its LIFO earnings relative to FIFO earnings. This phenomenon is called LIFO liquidation.
- During a fiscal period, $\Delta LIFO \text{ Reserve}$ captures the difference between $COGS_{LIFO}$ and $COGS_{FIFO}$, while the amount of LIFO Reserve at any point in time captures the *cumulative difference* between $COGS_{LIFO}$ and $COGS_{FIFO}$ since the time the company began using LIFO costing.

LIFO and the Tax Conformity Rule

- Firms using the LIFO costing method for tax purposes must also use LIFO for US GAAP purposes.
 - This is known as the **LIFO tax conformity rule**.
 - When prices are rising and a firm is increasing its LIFO reserve, LIFO results in a higher COGS and therefore a lower reported taxable income than either FIFO or the weighted average cost flow methods.
 - However, if LIFO reserves are decreasing, so that the firm suffers a LIFO liquidation, its COGS may be lower and therefore may report a higher taxable income than under the FIFO or weighted average cost flow methods.

Provisioning for Inventory Impairment

- GAAP requires firms to evaluate inventory for impairment by estimating the net realizable value of inventory. If the net realizable value is lower than cost, then inventory is written down. However, if the net realizable value is greater than inventory cost, the inventory *is not* written up. This rule is known as the lower of cost or net realizable value rule (LCNRV) for inventories.

COGS	\$x	
Allowance to Reduce Inventory to Net Realizable Value (Contra Asset)		\$x

Excerpt from Under Armour's Fiscal 2023 Annual Report:

Inventory Valuation and Reserves

Inventories consist primarily of finished goods. Costs of finished goods inventories include all costs incurred to bring inventory to its current condition, including inbound freight, duties and other costs. We value our inventory at standard cost which approximates landed cost, using the first-in, first-out method of cost determination. Net realizable value is estimated based upon assumptions made about future demand and retail market conditions.

If we determine that the estimated net realizable value of our inventory is less than the carrying value of such inventory, we record a charge to cost of goods sold to reflect the lower of cost or net realizable value. If actual market conditions are less favorable than those that we projected, further adjustments may be required that would increase the cost of goods sold in the period in which such a determination was made. As of March 31, 2023 and 2022, the inventory reserve was \$34.8 million and \$26.8 million, respectively.

LCNRV Rule: US GAAP vs. IFRS Difference

- Under US GAAP, inventory that has been written down cannot be later revalued to higher levels, even if the market value of that inventory has increased.
- However, IFRS does allow companies to reverse previous write-downs in an allowance account, so long as those reversals do not exceed the cumulative amount of prior write-downs. The reversal results in a debit to the allowance account and a credit to COGS.

Inventory Write-downs and Earnings Quality

- Note that one measure of earnings quality is the extent to which companies appropriately use the LCNRV rule to write down inventories.
 - Writing down inventory reduces gross margins, so companies may strategically delay recognizing inventory write-downs.
 - Excessive write-downs may also be used to boost profits in future periods at the expense of current-period profits.
 - Analyzing changes in the inventory reserve account (i.e., the allowance account) is therefore useful in gauging the quality of a firm's earnings.

Accrued Liabilities

- Reflect expenses that have been incurred during the period but not yet settled.
- Often referred to as *Accrued Expenses*, considering that the debit side of the accrual entry is reflected as an expense in the income statement.
- Common Accrued Expenses or Liabilities:
 - Wages, Salaries
 - Income Taxes, Rent
 - Utilities, Insurance
 - *Warranty Costs*
 - *Restructuring Costs*

} *Contingent liabilities*

Contingent Liabilities

- Some liabilities are less certain because the ultimate settlement of the liability is **contingent** on the outcome of a future event
- Examples of contingent liabilities include
 - Guarantees on the debt of another entity
 - Lawsuits
 - Product warranties and recalls
 - Environmental disasters and remediation
- Companies record a contingent liability when two conditions are met
 - It is **probable** that one or more future events will confirm that a liability existed at the financial statement date
 - The amount required to settle the liability in the future can be **reasonably determined** at the financial statement date

Warranty Liabilities

- Warranty liabilities are commitments that manufacturers make to repair or replace defective products within a specified period of time
- If the obligation is **probable** and the amount **estimable** with reasonable certainty, GAAP requires that companies
 - Record the expected cost of warranties as a **liability**
 - Record the related expected warranty **expense** in the income statement in the same period that the sales revenue is reported
- When the defective product is later replaced (or repaired), the **liability is reduced**, and the company records the cost of necessary to satisfy the claim

Product Warranties

- Product warranties provide for the repair or replacement of defective products and *may* or *may not* be valued separately from the transaction price of a product.
 - If the value of the warranty *can* be reliably measured and separated from the price of the product, such as for ***extended warranties***, then the warranty is treated as a *separate performance obligation*. The firm will then recognize ***unearned or deferred warranty revenue*** for the product at the time of sale and will recognize revenue over time.
 - But for ***assurance or limited warranties*** whose values cannot be separated from the price of the product, the accounting for warranty liability is treated as a contingent liability.

Harley's Warranty Liabilities

	2023	2022	2021
Balance, beginning of period	\$ 75,960	\$ 61,621	\$ 69,208
Warranties issued during the period	45,374	39,466	41,489
Settlements made during the period	(67,084)	(38,173)	(40,015)
Recalls and changes to pre-existing warranty liabilities	9,894	13,046	(9,061)
Balance, end of period	<u>\$ 64,144</u>	<u>\$ 75,960</u>	<u>\$ 61,621</u>

- At the beginning of fiscal 2023, Harley-Davidson reported a warranty and recall liability (reserve) of \$75,960 and \$64,144 at the end of fiscal 2023.
- During 2023 Harley-Davidson:
 - Added \$45,374 for warranties issued on products sold in 2022
 - Added \$9,894 for recalls and changes to pre-existing warranty liabilities
 - Paid out \$67,084 to settle warranty claims where cost of settlements include cash paid to customers for refunds, wages paid for repairs, and the cost of parts used in repairs

Restructuring Costs

- US GAAP and IFRS require firms to estimate the costs of restructuring activities (such as exit or disposal plans) and record this estimate as an expense, with an associated restructuring liability or provision.
 - To recognize a liability and the related expense, the restructuring activity must obligate the entity to transfer or use assets—that is, when an event leaves the company with little or no discretion to avoid transferring or using the assets in the future.
 - Note, therefore, that a *commitment* to a restructuring plan only expresses management's intended future actions and therefore does not meet the requirement for recognizing a liability and related expense.

Under Armour: Restructuring Plan

During Fiscal 2020, the Company's Board of Directors approved a restructuring plan ranging between \$550 million to \$600 million in costs (the "2020 restructuring plan") designed to rebalance the Company's cost base to further improve profitability and cash flow generation. The Company concluded the 2020 restructuring plan during the Transition Period.

Restructuring and related impairment charges and recoveries require the Company to make certain judgments and estimates regarding the amount and timing as to when these charges or recoveries occur. The estimated liability could change subsequent to its recognition, requiring adjustments to the expense and the liability recorded. On a quarterly basis, the Company conducts an evaluation of the related liabilities and expenses and revises its assumptions and estimates as appropriate, as new or updated information becomes available. No adjustments to expense were recorded during Fiscal 2023.

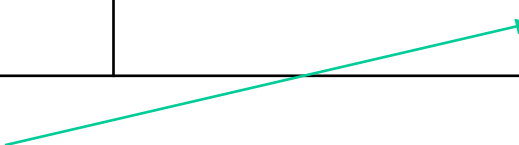
Under Armour: Restructuring Plan

	Employee Related Costs	Contract Exit Costs	Other Restructuring Related Costs
Balance as of December 31, 2020	\$ 12,868	\$ 61,642	\$ 6,098
Net additions (recoveries) charged to expense	(1,655)	17,814	(1,494)
Cash payments	(5,473)	(47,486)	(6,078)
Foreign exchange and other	(2,192)	(565)	120
Balance as of December 31, 2021	<u>\$ 3,548</u>	<u>\$ 31,405</u>	<u>\$ (1,354)</u>
Net additions (recoveries) charged to expense	(10)	58,555	(1,871)
Cash payments	(955)	(9,280)	—
Foreign exchange and other	89	(2,443)	3,225
Balance as of March 31, 2022	<u>\$ 2,672</u>	<u>\$ 78,237</u>	<u>\$ —</u>
Net additions (recoveries) charged to expense	—	—	—
Cash payments	(1,057)	(76,287)	—
Foreign exchange and other	(659)	(1,832)	—
Balance as of March 31, 2023	<u>\$ 956</u>	<u>\$ 118</u>	<u>\$ —</u>

Provisions and Earnings Quality

Provisions (Accrued Warranty or Restructuring Liabilities)

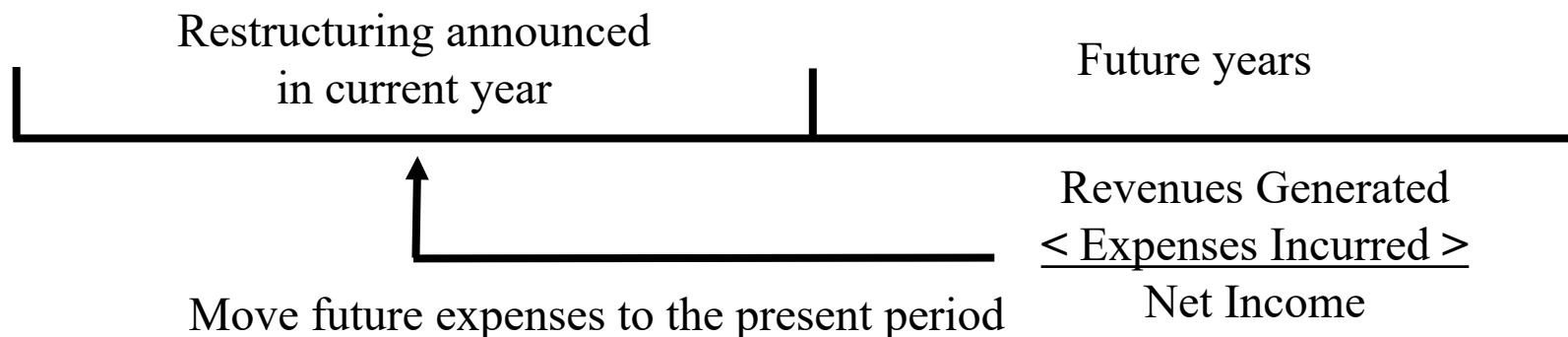
Actual Amount	BB
	Provision (estimate) for year
	EB (estimate)



The ending balance is estimated and is therefore discretionary. This affects the amount provisioned (expense recognized) during a fiscal period.

- By understating or overstating the ending balance in a reserve, management may manipulate net income by understating or overstating the amount of expenses recognized during a period.

Cookie Jar Reserves and Earnings Quality



- Suppose a firm announces a long-term restructuring plan to lay off employees with a severance package expected to cost \$50 million in fiscal year 2025:

Restructuring Expense (SG&A) \$50 million

Restructuring Liability \$50 million

- Over the next two fiscal years, 2025 and 2026, employees were terminated resulting in total cash payoffs of \$25 million:

Restructuring Liability \$25 million

Cash \$25 million

- In future years, i.e., fiscal years 2027 and after, firm can understate other operating expenses of say \$Y million by gradually reversing the unpaid reserve:

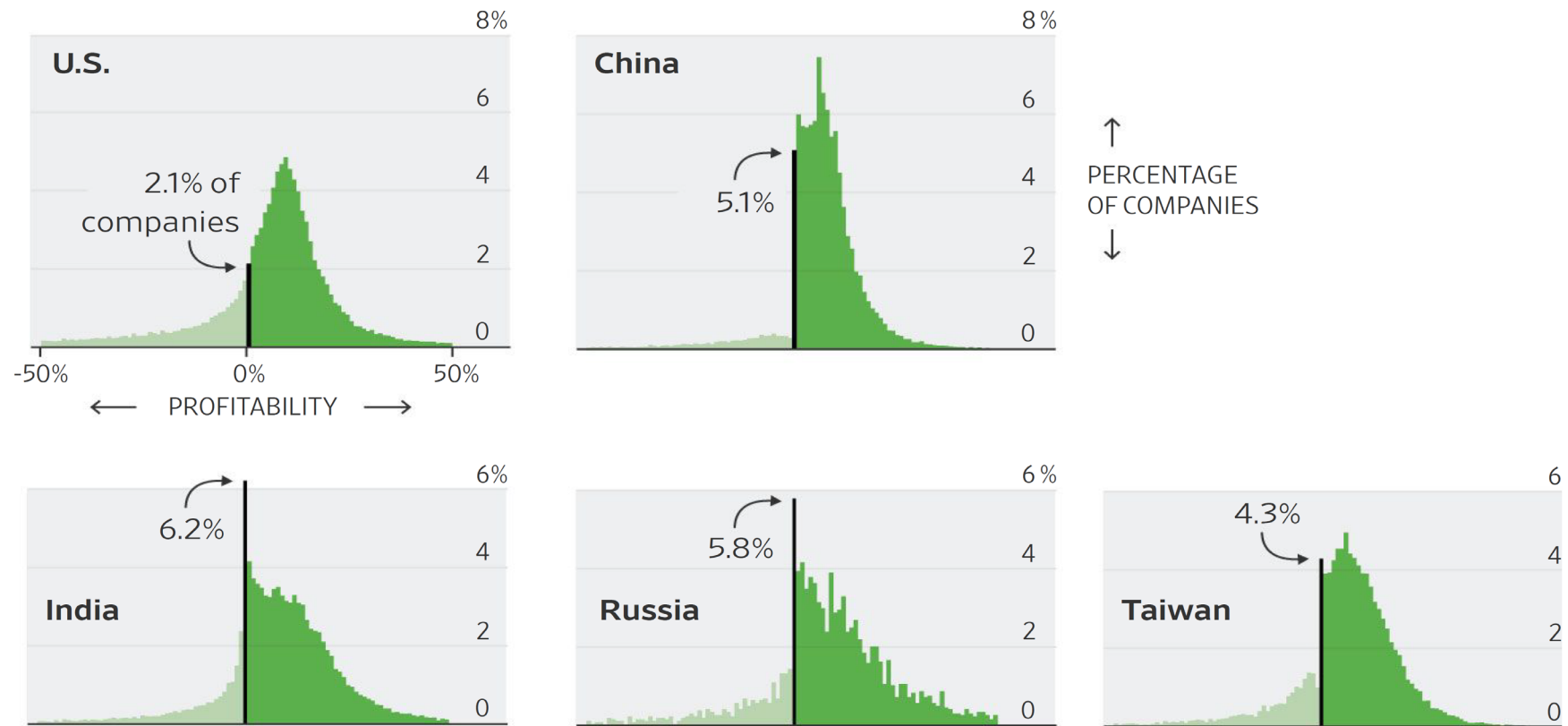
Restructuring Liability \$ y million

Cash \$ y million

Kinky Earnings: Earnings Management?

Publicly traded companies

■ Unprofitable ■ Profitable ■ Reported 0 to 1% profitability



Note: Profitability measured by return on equity, or reported net income as percent of average book value of shareholders' equity, 1998-2018.

Source: Rayliant Global Advisors

Contingent Liabilities

- GAAP requires that contingent liabilities reflect **estimated cost** the company expects to incur as a result of such liabilities.
- Companies might intentionally
 - **Underestimate** contingent liability to report **higher current income**.
 - **Overestimate** contingent liability to **depress current income** and create a liability (cookie jar reserve) to absorb future costs.
- These actions affect income in both the current and subsequent periods – both periods are incorrect.
- We must closely examine contingent liabilities to identify earnings manipulations.

Long-Lived Operating Assets

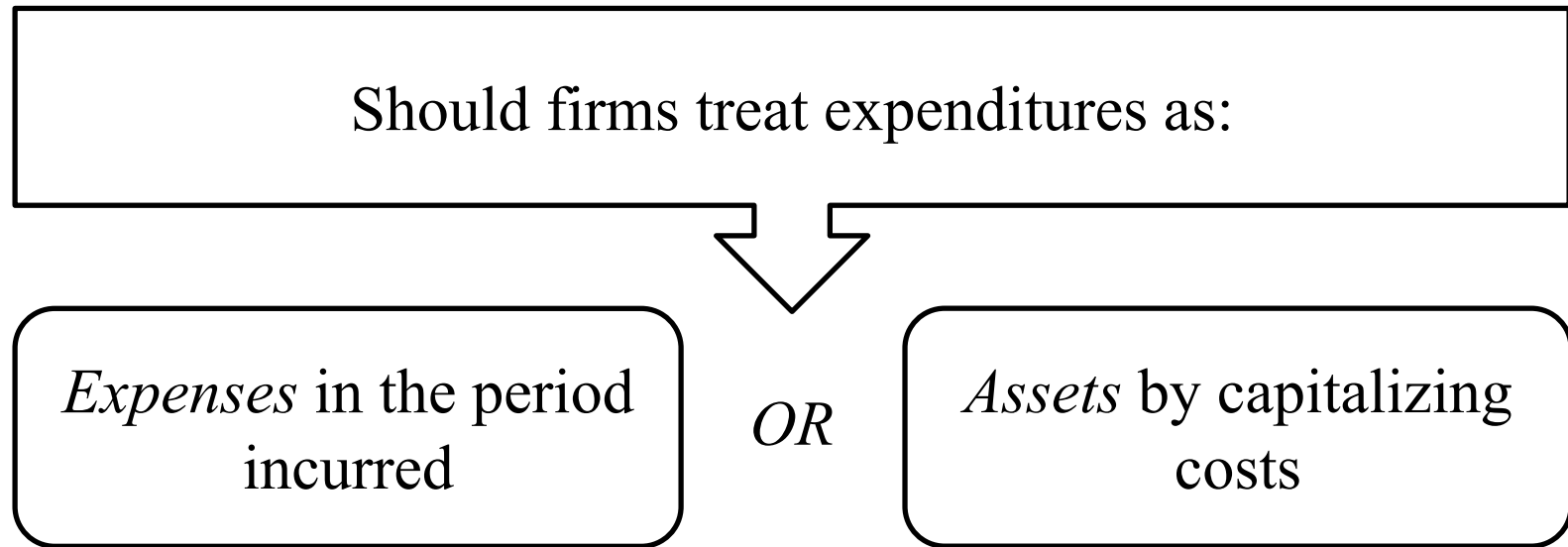
- Category 1: PP&E, tangible assets with definite lives and Land.
- Category 2: Intangible Assets with definite and indefinite lives.
- Category 3: Right-of-Use Assets (we will study these later).
- Category 4: Goodwill.
 - Note that all intangibles in categories 2 and 3 are sometimes called *identifiable* intangibles to distinguish them from goodwill, which is *not* an identifiable asset as goodwill cannot be valued separately from a business. We will also study goodwill later in the course.

Long Lived Assets: LVMH

CONSOLIDATED BALANCE SHEET

Assets	Notes	June 30, 2024	Dec. 31, 2023	June 30, 2023
<i>(EUR millions)</i>				
Brands and other intangible assets	3	25,895	25,589	25,319
Goodwill	4	21,379	24,022	25,102
Property, plant and equipment	6	28,531	27,331	24,971
Right-of-use assets	7	16,060	15,679	14,642
Investments in joint ventures and associates	8	1,388	991	1,074
Non-current available for sale financial assets	9	1,146	1,363	1,394
Other non-current assets	10	1,032	1,017	1,102
Deferred tax		4,094	3,992	3,769
Non-current assets		99,525	99,984	97,373

Capitalizing vs. Immediate Expensing of Expenditures



- An ***expenditure*** is *capitalized* (and recognized as an asset) if the firm:
 1. Has acquired rights to future use as the result of a past transaction, and
 2. Can measure or quantify the future benefits.

Treatment of Expenditures with Potential Long-Term Benefits

Nature of Resource		
	Tangible	Intangible
Acquired Internally	Self-constructed Buildings and Equipment (asset)	Research and Development (expense under U.S. GAAP; research is an expense and development is an asset under IFRS) Advertising (expense) Employee Training (expense) Software Development Costs: Pre-technological Feasibility (expense) Post-technological Feasibility (asset)
Acquired Externally	Land, Buildings, and Equipment (asset)	Proved Technologies (asset) In-Process Technologies (asset) Patents, Trademarks, Customer Lists, and Other Identifiable Resources (asset) Trained Labor Force and Other Unidentifiable Resources (part of goodwill) Goodwill (asset)

Most intangible assets on a balance sheet are created in acquisitions.

Measurement of Acquisition Cost

- The acquisition value includes *all costs of acquiring and putting the asset into use*:
 - Purchase price (less discounts)
 - Transportation and handling
 - Insurance costs during shipping
 - Installation/customization/inspection costs
 - Borrowing costs (capitalized interest) if self-constructed

Costs Subsequent to Acquisition

- Additional costs incurred after an asset is placed in service:
 - Improvement or betterment, consisting of financial outlays that enhance the usefulness of the asset or extend the asset's useful life beyond the original expectation
 - Capitalized as part of asset.
 - Routine repairs and maintenance
 - Expensed in the period incurred

Intangible Assets

- Assets that may provide future benefits without having physical form; such as patents, copyrights, customer lists, etc.
 - Such assets are called *intangible assets*.
- Two types of Intangible Assets: internally developed vs. purchased:
 1. *Internally Developed:*
 - Examples: R&D, Advertising (brand value), Training (human capital)
 - Accounting treatment: costs are expensed immediately
 2. *Purchased (directly or via business combination)*
 - Examples: Patents, Trademarks, In-Process R&D, Goodwill
 - Accounting treatment: costs are capitalized as an asset and amortized and/or impaired over time.

Treatment of Acquisition Cost

- If a firm *consumes* the asset while generating revenues (e.g., a machine or patent):
 - Allocate some cost to each period (match expense to the benefit)
 - For tangible assets, this cost allocation is called *depreciation*, while for definite-lived intangible assets, the cost allocation is called *amortization*.
- If a firm does *not consume* the asset while generating revenues (e.g., land):
 - Do not allocate expenses (maintain at acquisition cost until asset is sold or is impaired)

Depreciation and Amortization

- Depreciation: allocating the cost of a *tangible* asset over its useful life.
- Amortization: allocating the cost of a definite-lived *intangible* asset over its useful life.
- Tangible assets (e.g., PP&E)

Depreciation Expense	\$x	
Accumulated Depreciation–PP&E		\$x

- Intangible assets with definite lives: (e.g., Patent)

Amortization Expense	\$y	
Accumulated Amortization–Patent		\$y

Depreciation Methods for PP&E

- **Methods used for depreciation PP&E with finite lives in financial reporting include:**
 1. Straight-line (time) method.
 2. Units-of-production method, straight line against production measure other than time.
 3. Accelerated methods.
 4. Decelerated methods (rare).

Amortization of Intangibles

- Definite-Lived Intangibles have limited economic lives and are therefore amortized over their economic lives, usually by a straight-line method:
 - Patents, customer contracts, software.
 - We will see later that definite-lived intangibles *may* also be tested for impairment under certain circumstances.
- However, Indefinite-Lived Intangibles are *not amortized* and are instead tested for impairment at least once a year:
 - In-Process R&D, Brand Name, Goodwill.

Change in Accounting Estimates

- When necessary, companies can change estimates as a result of obtaining new information. Revisions in estimates are handled currently and *prospectively*:
 - e.g., change in useful life and/or salvage value of assets, inventory obsolescence, depreciation method, etc.
- At the beginning of fiscal 2020, Dodo Corp. (Dodo) purchased a building for \$300,000 and estimated its useful life to be 15 years with no residual value. It depreciated the building on a straight-line basis.
 - After 5 years of service, i.e., at the beginning of fiscal 2025, Dodo acquired new information that led it to revise its estimate of the building's original useful life downward to 10 years, instead of 15 years. What is the carrying value of the building at the end of fiscal 2025?

Change in Estimate

Buildings—Gross	\$300,000
Less Accumulated Depreciation $[(300,000 - 0)/15]*5$	100,000
Building—Net, beginning of fiscal 2025	\$200,000

- During fiscal 2025, Dodo would depreciate the building using the depreciable base of \$200,000 over the *new remaining life of 5 years*:

Depreciation Expense = $$(200,000 - 0)/10-5$	\$40,000
Accumulated Depreciation	\$40,000

- Carrying value of Building at end of fiscal 2025, would be:

Buildings—Gross	\$300,000
Less Accumulated Depreciation [= 100,000 + 40,000]	140,000
Building—Net, end of fiscal 2025	\$160,000

Change in Estimates vs. Correction of Errors

- Unlike changes in estimates that *do not* affect prior-period financials, corrections of errors/intentional misstatements require the restatement of prior financial statements and should be treated *retrospectively*. As soon as a company discovers a misstatement, it must correct it.
- Companies record such corrections from prior periods as an adjustment to the *beginning balance of retained earnings* in the current period. Such corrections are called *prior period adjustments*.

Correction of Error: Example

At the beginning of fiscal 2020, Dodo Corp. (Dodo) purchased a building for \$300,000 and estimated its useful life to be 25 years with no residual value. It depreciated the building on a straight-line basis. After 5 years of service, at the beginning of fiscal 2025, Dodo's auditor discovered that the company had incorrectly depreciated the building based on a useful life of 15 years instead of 25 years. Assuming that Dodo owed taxes for the past five years and paid them in cash each year at a flat tax rate of 20%, how should Dodo Corp. account for this error in fiscal 2025?

Correction of Errors

Incorrect Treatment for each fiscal year 2020 through 2024

$$\text{Depreciation Expense} = \$20,000 = (\$300,000 - 0) / 15$$

$$\text{Accumulated Depreciation} \quad \$20,000$$

$$\text{Cash (tax savings of } \$20,000 * 0.20) = \$4,000$$

$$\text{Income tax Benefit} \quad \$4,000$$

Impact on Net Income for each fiscal year 2020 through 2024

$$\begin{aligned} \text{Depreciation Expense} &= \$ -20,000 \\ + \text{Income Tax Benefit} &= \quad 4,000 \\ \hline \text{Impact on Net Income} &= \$ -16,000 \end{aligned}$$

Correct Treatment for each fiscal year 2020 through 2024

$$\text{Depreciation Expense} = \$12,000 = (\$300,000 - 0) / 25$$

$$\text{Accumulated Depreciation} \quad \$12,000$$

$$\text{Cash (tax savings of } \$12,000 * 0.20) = \$2,400$$

$$\text{Income Tax Benefit} \quad \$2,400$$

$$\begin{aligned} \text{Depreciation Expense} &= \$ -12,000 \\ + \text{Income Tax Benefit} &= \quad 2,400 \\ \hline \text{Impact on Net Income} &= \quad \$ -9,600 \end{aligned}$$

Correction of Errors

- For *each* fiscal year 2020 through 2024, net income of Dodo was therefore *understated* by $\$16,000 - 9,600 = \$6,400$, taxes owed were *understated* by $\$4,000 - 2,400 = \$1,600$, and Accumulated Depreciation was *overstated* by $\$20,000 - 12,000 = \$8,000$. To correct the balance sheet for the errors in the prior five fiscal years, at the beginning of fiscal 2025, the following correcting journal entry should be made:

Correcting entry to restate B/S during fiscal 2025:

Accumulated Depreciation	$\$40,000 = \$8,000 * 5$
Retained Earnings	$\$32,000 = \$6,400 * 5$
Taxes Payable	$8,000 = 1,600 * 5$

Reviewing PP&E and A/D Accounts (Ignoring Impairments and Revaluations)

The diagram illustrates the flow of PP&E transactions between four accounts:

- PP&E (Asset) Account:**
 - BB (Beginning Balance):** Purchases (debit side), Sales (original cost) (credit side).
 - EB (Ending Balance):** Cash used, if any, to purchase PP&E (debit side).
- Accumulated Depreciation (A/D) (contra-asset) Account:**
 - BB (Beginning Balance):** Increase in A/D due to depreciation expense recognized on existing PPE during period (debit side).
 - EB (Ending Balance):** Carrying Value (CV) = Original Cost - A/D (credit side).
- Cash (Asset) Account:**
 - BB (Beginning Balance):** Cash received, if any, from sale of PP&E (debit side).
 - EB (Ending Balance):** Cash used, if any, to purchase PP&E (credit side).
- Income Statement (I/S):**
 - Depreciation Expense (I/S):** Depreciation expense recognized during period (debit side).
 - Gain/Loss from PP&E Sale (=cash received from sale less CV) (I/S):**
 - Loss, if any, on PP&E sold (debit side).
 - Gain, if any, on PP&E sold (credit side).

Flow of Transactions (Arrows):

- (1) Purchases:** From PP&E (Asset) BB to Cash (Asset) BB.
- (2) Depreciation Expense:** From Accumulated Depreciation (A/D) BB to Depreciation Expense (I/S).
- (3) Sales:** From PP&E (Asset) BB to Cash (Asset) BB, and from Accumulated Depreciation (A/D) BB to Gain/Loss from PP&E Sale (I/S).

Entries to Capture Movements in PPE Related Accounts

1. *Purchase of Equipment for cash:*

Equipment		
Cash		

2. *Recognition of Depreciation Expense during the year:*

Depreciation Expense		
Accumulated Depreciation		

3. *Sale of Equipment during the year:*

	Accumulated Depreciation		
	Cash		
	Loss on Sale		
<i>Either Gain or Loss</i>	Equipment–Gross		
	Gain on Sale		

